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REMOTE SENSING AND BIG GEODATA
EXERCISE 6
MACHINE LEARNING

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Contents

Introduction 3
Supervised Learning 3
Unsupervised Learning 7
Conclusion 10

Introduction

Machine Learning (ML) is the branch of artificial intelligence that enables systems to learn autonomously from data, without being explicitly programmed and without requiring prior domain expertise. The goal is for machines to learn from data and generate accurate predictions.
Unlike traditional programming, where inputs and rules together produce an output, ML works the other way around: inputs and outputs are paired together so the system can figure out the rules itself.
Within ML, we can distinguish two different families of algorithms: supervised and unsupervised.

- In supervised learning, the model is trained on a labelled dataset (training set), from which it learns the relationship between pixel spectral values and land cover classes.
- In unsupervised learning, by contrast, no labels are provided: the algorithm independently identifies groups of pixels with similar spectral characteristics, producing clusters that the operator then interprets and assigns to real world classes at a later stage.

In the context of remote sensing, ML offers the ability to handle high-dimensional data and to map classes with highly complex spectral signatures.

For this exercise, we carried out both types of classification in Google Earth Engine. The first exercise involves supervised classification using a Support Vector Machine (SVM) applied to a Landsat 8 composite, with the goal of distinguishing between two land cover classes.
The second exercise instead addresses unsupervised classification through the K-means algorithm applied to a Landsat 7 composite, aiming to identify five spectrally homogeneous clusters without any prior knowledge of the classes.

Supervised Learning

Supervised learning is a machine learning approach in which the model is built from a set of labelled examples known as training data; the algorithm optimizes its internal parameters to minimize classification error on the training set, acquiring in this way the ability to correctly associate a pixel's spectral signature with a land cover class.
The typical supervised classification workflow in GEE follows four main steps:

- First, we acquire and preprocess the satellite image by filtering it by date, area of interest, and cloud cover, ultimately computing a median composite;
- Then we have to define the training areas, polygons drawn manually on the image that serve as reliable samples for each class.
- From these polygons, spectral values of the corresponding pixels are extracted using “sampleRegions”, allowing us to build the training dataset.
- Finally, we train the classifier and apply it to the entire image to produce the classified map.

The Support Vector Machine is a classification algorithm that separates data using a linear decision surface (hyperplane) that maximize the margin between the two classes. This makes it ideal for single class classification tasks, such as the one we carried out here.

I chose to conduct this exercise over an area of the Bolivian rainforest, a geographically significant setting for environmental monitoring, given that Bolivia ranks among the countries with the highest tropical deforestation rates in recent years.
The goal was to produce a binary classification capable of distinguishing intact forest pixels from deforested or bare soil pixels.

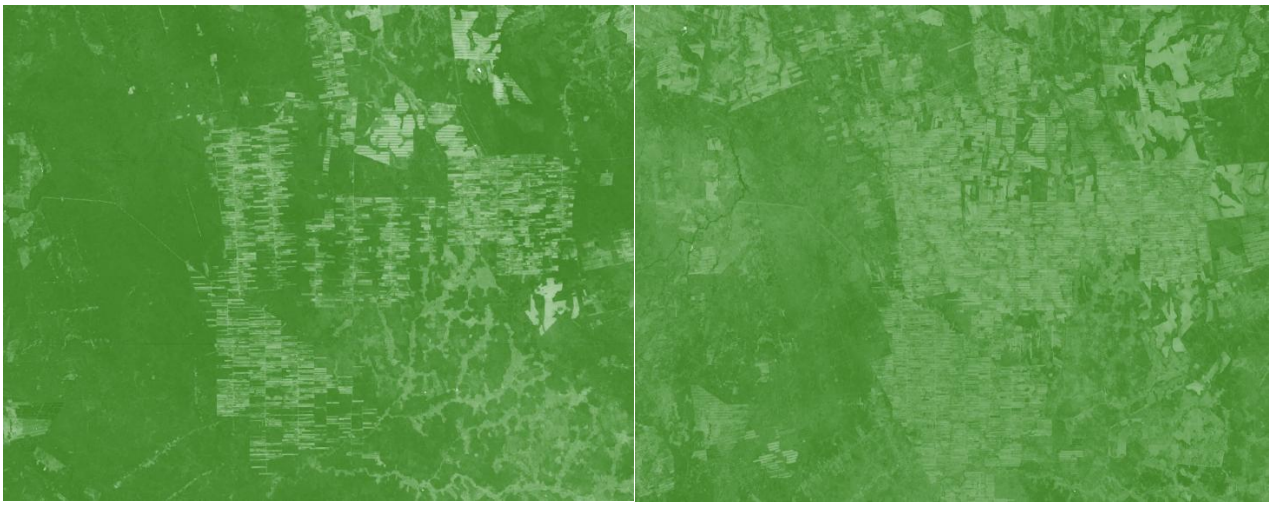


Figure 1: NDVI 2018 - 2025

Comparing the two NDVI composites reveals a clear progression of deforestation between 2018 and 2025. In 2018, forest cover appears dominant, with high vegetation values extending across most of the area; however, some linear clearings are already visible, consistent with the characteristic herringbone pattern typical of agricultural expansion in the Bolivian Amazon.
By 2025, the landscape is noticeably more fragmented, with a marked spread of lower NDVI areas indicating a reduction in canopy biomass and a strong advance of the deforestation frontier.

To assess the extent of deforestation in the study area, we implemented a supervised machine learning approach in Google Earth Engine using Landsat 8 Level-2 imagery.
The image composite was generated for summer 2025 by filtering scenes according to the study area boundaries and a maximum cloud cover threshold of 50%.

For the SVM classification, we manually defined four training polygons, two labelled as class 0 (NoForest, NoForest2) and two as class 1 (Forest, Forest2). These polygons were merged into a “FeatureCollection” with a "class" property assigned to each feature and used as input for the “sampleRegions()” function, which extracts the spectral values of all pixels falling within the polygons across the selected bands, constructing the training dataset.



Figure 2: training polygons

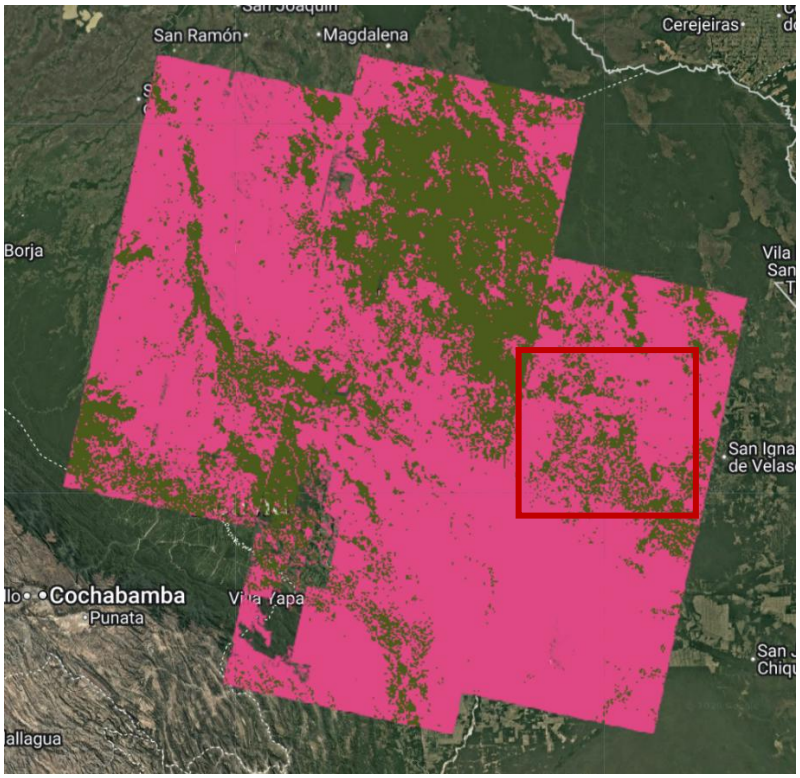
Spectral informations from all Landsat visible, near-infrared, shortwave-infrared and themral bands were extracted and used to train the Support Vector Machine (SVM) classifier with a Radial Basis Function (RBF) kernel.

```
45
46  //SUPERVISED CLASSIFICATION
47  var polygons = ee.FeatureCollection([
48    ee.Feature(NoForest, {'class':0}),
49    ee.Feature(NoForest2, {'class':0}),
50    ee.Feature(Forest, {'class':1}),
51    ee.Feature(Forest2, {'class':1}),
52  ])
53  print(polygons)
54  Map.addLayer(polygons)
55
56  //training
57  var training_set2025 = L8_2025_scaled_median.sampleRegions({
58    scale:30,
59    collection: polygons,
60    properties:['class'],
61    geometries:true
62  })
63  print(training_set2025.limit(10))
64  Map.addLayer(training_set2025)
65
66  //classifier (SVM)
67  var classifier = ee.Classifier.libsvm({
68  kernelType: 'RBF',
69    gamma: 0.5,
70    cost: 10
71  })
72  print(classifier)
73
74  //we have to train our classifier
75  var bands = ['SR_B1', 'SR_B2', 'SR_B3', 'SR_B4', 'SR_B5', 'SR_B6', 'SR_B7', 'ST_B10'];
76  var train_classifier = classifier.train(training_set2025,'class', bands)
77  //                          imput data,            algorithm,     what we want to predict, bands
78  print(train_classifier)
79
80  //creation of the final image
81  var class_image = L8_2025_scaled_median.classify(train_classifier)
82  Map.addLayer(class_image.randomVisualizer(),{},'2025 Deforestation')
83  //pink areas = deforestazione
```

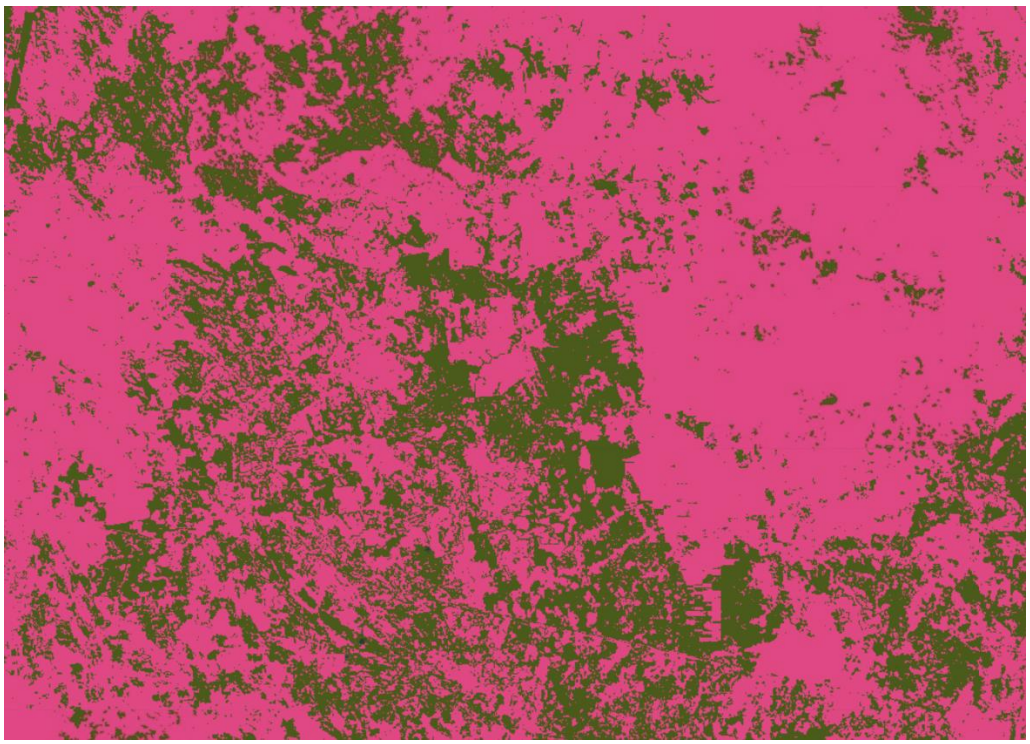
Figura 3: GEE code for the supervised classification

The classification results clearly highlight extensive deforestation across the study area. The resulting map shows a large concentration of non-forest areas (displayed in pink) distributed in a distinctive linear and rectangular pattern, characteristic of agricultural expansion and land conversion activities. These geometric patterns contrast sharply with the remaining forested areas (green), which are concentrated in irregular patches, predominantly along the upper-right portion of the image.

Deforested areas are particularly extensive in the central portions of the scene and in the upper-left and lower-right corners of the study area; they are not placed randomly, but follow a fragmented pattern that reflects the advancing agricultural frontier.



Zooming into the central-right portion of the area, as seen in the NDVI maps:



Overall, the supervised classification, when compared against the underlying satellite layers and the NDVI maps, successfully identified the spatial distribution of deforested land and revealed a highly fragmented landscape. The observed patterns are consistent with large-scale forest conversion processes occurring in eastern Bolivia, where, as said before, agricultural expansion and land-use change represent the primary drivers of deforestation.

Unsupervised Learning

Unsupervised classification belongs to a family of ML algorithms in which the model is provided with no prior labels. Rather than learning the correspondence between spectral signatures and classes, the algorithm autonomously seeks out latent structure in the data, grouping pixels according to their spectral similarity.
The result is a set of clusters, groups of pixels sharing similar characteristics, which we must then interpret and associate with land cover categories at a later stage, drawing on prior knowledge or reference data.
The K-means algorithm is the most widely used clustering method in remote sensing image analysis. It operates iteratively: given a desired number K of clusters, the algorithm creates centroids in feature space (randomly) assigns each pixel to its nearest centroid, recomputes each centroid as the mean of the pixels assigned to it, and repeats the last two steps until cluster assignments no longer change significantly.

For this second exercise, I've chosen the island of Mallorca, Spain, as the study area, using Landsat 7 Level 2, Collection 2, Tier 1 imagery filtered over the time window from August 1st to December 31st, 2020, with an additional cloud cover filter excluding scenes with more than 50% cloud coverage.
The goal, as previously described, was to automatically classify the main land cover types present on the island into four classes. As a first step, I extracted a random sample of 5,000 pixels from the median composite within the area of interest, at a spatial scale of 30 meters consistent with Landsat's native resolution.
These pixels, along with their multispectral signatures, formed the training set on which the clusterer was then fitted.

```
42  //UNSUPERVISED CLASSIFICATION
43  //training set
44  var training_set = L7_median.sample({
45    region:geometry,
46    scale:30,
47    //numer of points
48    numPixels:5000,
49    geometries:true
50  })
51  print(training_set)
52  Map.addLayer(training_set)
```

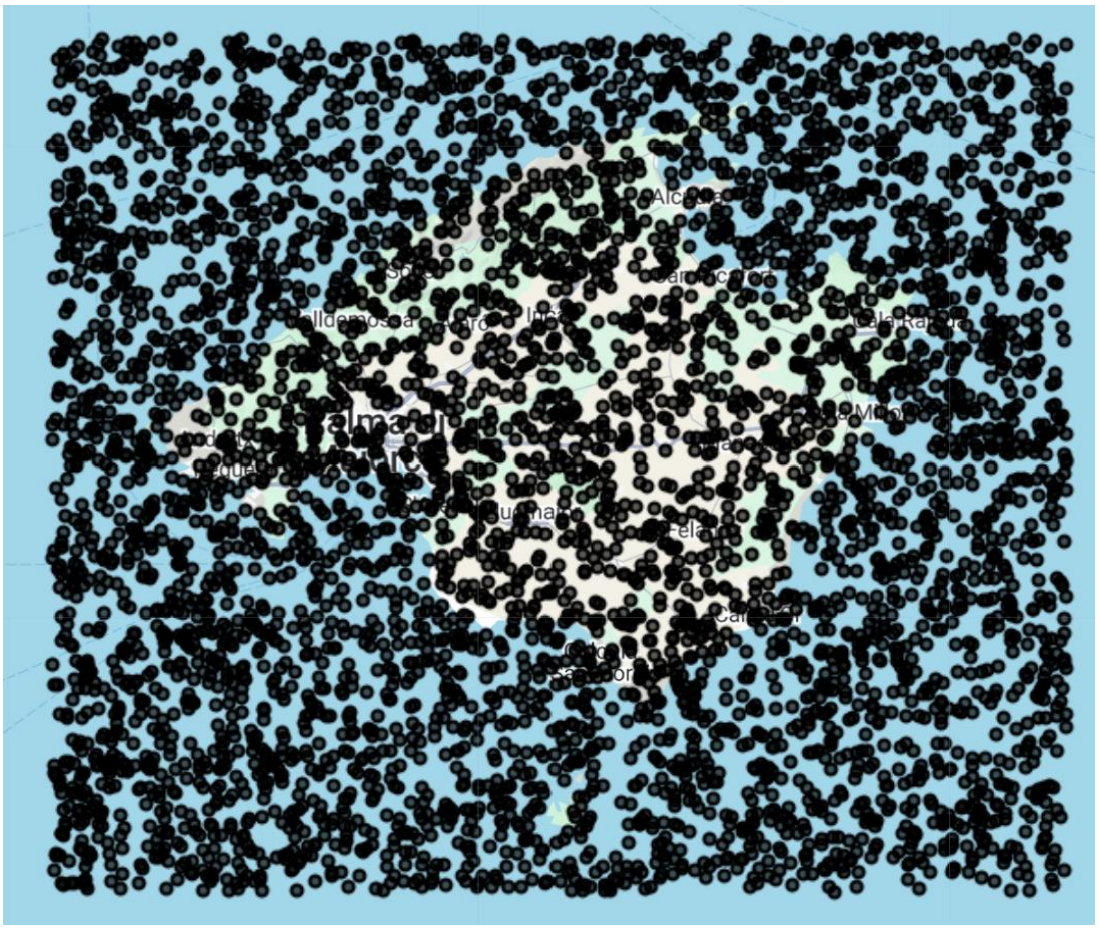


Figura 4: training set

Once training was complete, the model was applied to the entire image, producing a classification map in which every pixel has been assigned to one of the four classes identified by the algorithm.

```
var number_clusters = 4
var clusterer=ee.Clusterer.wekaCascadeKMeans(number_clusters)

//alleniamo
var trained_clusterer = clusterer.train(training_set)
print(trained_clusterer)
//applichiamolo all'immagine
var clustered_L7=L7_median.cluster(trained_clusterer)

Map.addLayer(clustered_L7.randomVisualizer(),{},'random')
//brown = agricultural areas
//green = water
//pink = urban areas
//yellow = vegetation
```

The final map shows a spatial distribution of classes consistent with the known morphology and land use of the island.

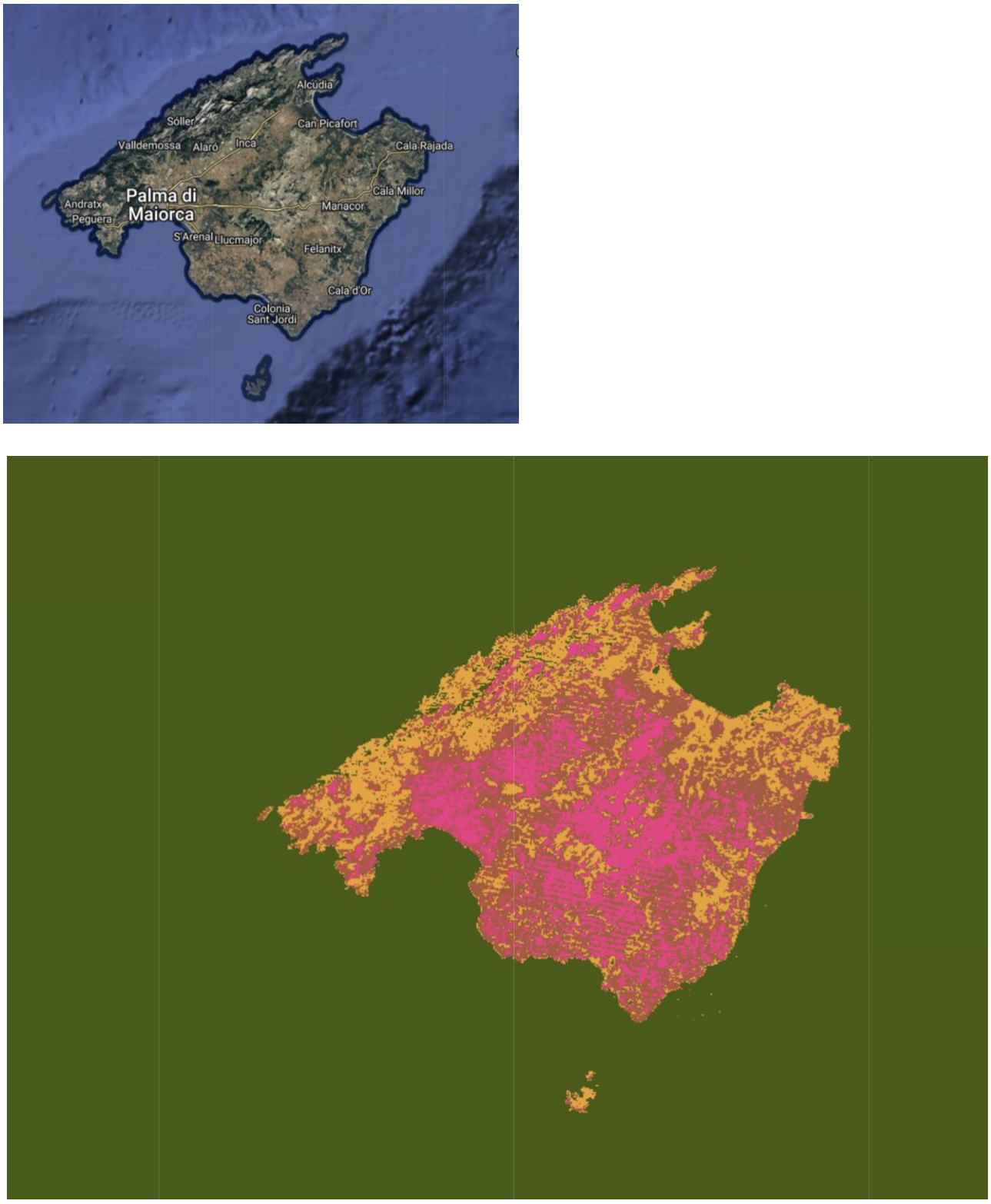


Figura 5: Unsupervised classification of Maiorca

By interpreting the classes based on their geographic distribution, I was able to associate each colour with the following land cover types:

- brown corresponds to agricultural areas,
- pink represents urban surfaces and bare soil,
- yellow identifies natural vegetation,
- green delineates water and the surrounding marine surfaces.

Conclusion

To conclude, this exercise has shown how Machine Learning techniques represent effective tools for the analysis and classification of satellite imagery in the field of remote sensing.
Through the supervised SVM approach, we were able to accurately identify areas affected by deforestation processes in the Bolivian rainforest, highlighting the substantial expansion of non-forest surfaces between 2018 and 2025 and the progressive fragmentation of the natural landscape.
At the same time, unsupervised classification via the K-means algorithm allowed us to identify and distinguish the main land cover types present on the island of Mallorca without relying on any labelled data.
Overall, the results confirm that the combination of multispectral satellite imagery, Google Earth Engine, and Machine Learning algorithms constitutes a robust and flexible approach for environmental monitoring and the analysis of land cover dynamics.